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A FLUX-BASED HDG METHOD

ISSEI OIKAWA

Abstract. In this paper, we present a flux-based formulation of the hybridizable discontinuous Galerkin (HDG) method for steady-state diffusion problems and propose a new method derived by letting a stabilization parameter tend to infinity. Assuming an inf-sup condition, we prove its well-posedness and error estimates of optimal order. We show that the inf-sup condition is satisfied by some triangular elements. Numerical results are also provided to support our theoretical results.

1. Introduction

We consider the hybridizable discontinuous Galerkin (HDG) method for the steady-state diffusion problem with Dirichlet boundary condition

$$\boldsymbol{q} + \nabla u = 0 \qquad \text{in } \Omega, \tag{1a}$$

$$\nabla \cdot \boldsymbol{q} = f \qquad \text{in } \Omega, \tag{1b}$$

$$u = 0 \qquad \text{on } \partial\Omega, \tag{1c}$$

where $\Omega \subset \mathbb{R}^d$ $(d = 2, 3)$ is a bounded convex polygonal or polyhedral domain and f is a given function. In the original HDG method [3], a numerical trace $\widehat{u}_h$ is introduced as an unknown variable to approximate the trace of u on element boundaries, which corresponds to a Dirichlet boundary condition, and a numerical flux $\widehat{\boldsymbol{q}}_h$ is properly defined. The other variables u_h and $\boldsymbol{q}_h$ approximating u and $\boldsymbol{q}$, respectively, can be eliminated in element-by-element fashion and we obtain a globally-coupled system of equations only in terms of $\widehat{u}_h$, which is called static condensation.

In [2], a flux-based formulation is presented, in which the trace of $\boldsymbol{q}$ on element boundaries instead of $\widehat{u}_h$ is hybridized, in other words, $\widehat{\boldsymbol{q}}_h$ is an unknown variable and $\widehat{u}_h$ is defined in terms of $\widehat{\boldsymbol{q}}_h$ and the other variables. The flux-based method is a rewrite of the original HDG method and provides the same solution, however, the local problem has a Neumann boundary condition, so that the static condensation is different from that of the original method. We note that the local solvability of the flux-based method is not obvious, which will be verified in Section 2.4.

In this paper, we propose a new flux-based method derived by passing the stabilization parameter to infinity. In our method, $\widehat{\boldsymbol{q}}_h$ is unknown and the numerical trace is defined by $\widehat{u}_h = u_h$. Since our method has saddle point structure, its well-posedness depends on whether the inf-sup condition we define in Section 2.3 is satisfied. The inf-sup condition is

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fulfilled if we use triangular meshes and the polynomials of degree k and $k+1$ for $\widehat{\boldsymbol{q}}_h$ and u_h, respectively, with a non-negative integer k. In addition, the proposed method using such approximation spaces achieves the optimal convergence rates in u_h and $\boldsymbol{q}_h$, like the HDG method with the so-called Lehrenfeld–Schöberl stabilization proposed in [5] and analyzed in [6, 7]. Although, in the Lehrenfeld–Schöberl stabilization, the L^2-orthogonal projection onto the approximation space of $\widehat{u}_h$ is inserted in front of u_h, such a projection is not used in our method because it is naturally incorporated through the transmission condition in a flux-based formulation.

The rest of the paper is organized as follows. In Section 2, we introduce notation and present the flux-based formulation, and a new method is derived from it. We verify the local solvability of the methods. In Section 3, we establish a priori estimate and error estimates of optimal order for our method, assuming that an inf-sup condition holds. In Section 4, we prove that the inf-sup condition is satisfied if the polynomial degrees for $\widehat{\boldsymbol{q}}_h$ and u_h are k and $k+1$, respectively, and triangular meshes are used. In Section 5, numerical results are presented to validate our theoretical results.

2. A flux-based HDG formulation

2.1. **Notation.** To begin with, we introduce notation to present the HDG method via flux hybridization. Let $\{\mathcal{T}_h\}_h$ be a family of meshes satisfying the quasi-uniform condition, where h stands for the mesh size. Let $\mathcal{E}_h$ denote the set of all edges or faces of elements in $\mathcal{T}_h$. Let $L^2(\mathcal{E}_h)$ denote the L^2-space on $\bigcup_{e\in\mathcal{E}_h} e$ and $P_m(\mathcal{T}_h)$ and $P_m(\mathcal{E}_h)$ denote the spaces of element-wise and edge-wise polynomials of degree m, respectively.

We use the usual symbols of Sobolev spaces [1], such as $H^m(D)$, $H^m(D)^d$, $\|\cdot\|_{m,D} := \|\cdot\|_{H^m(D)}$, and $|\cdot|_{m,D} := |\cdot|_{H^m(D)}$ for a domain D and an integer m. We may omit the subscripts when $D=\Omega$ or $m=0$, such as $\|\cdot\|_m = \|\cdot\|_{m,\Omega}$, $\|\cdot\| = \|\cdot\|_{0,\Omega}$, and $|\cdot|_m = |\cdot|_{m,\Omega}$. The piecewise Sobolev space of order m is denoted by $H^m(\mathcal{T}_h)$. The inner products are defined as

$$
\begin{aligned}
(\boldsymbol{q},\boldsymbol{v})_K &= \int_K \boldsymbol{q}\cdot\boldsymbol{v}dx, \quad (\boldsymbol{q},\boldsymbol{v})_{\mathcal{T}_h} = \sum_{K\in\mathcal{T}_h}(\boldsymbol{q},\boldsymbol{v})_K,\\
(u,w)_K &= \int_K uwdx, \quad (u,w)_{\mathcal{T}_h} = \sum_{K\in\mathcal{T}_h}(u,w)_K,\\
\langle u,w\rangle_{\partial K} &= \int_{\partial K} uwds, \quad \langle u,w\rangle_{\partial\mathcal{T}_h} = \sum_{K\in\mathcal{T}_h}\langle u,w\rangle_{\partial K},\\
\langle \mu,\lambda\rangle_{\mathcal{E}_h} &= \sum_{e\in\mathcal{E}_h}\int_e \mu\lambda ds, \quad (u,w) = \int_\Omega uwdx.
\end{aligned}
$$

We define the induced norms from these inner products by

$$
\|\boldsymbol{v}\|_{\mathcal{T}_h} = (\boldsymbol{v},\boldsymbol{v})_{\mathcal{T}_h}^{1/2}, \quad \|w\|_{\mathcal{T}_h} = (w,w)_{\mathcal{T}_h}^{1/2}, \quad \|w\|_{\partial\mathcal{T}_h} = \langle w,w\rangle_{\partial\mathcal{T}_h}^{1/2}.
$$

Throughout the paper, we use the symbol C to denote a generic constant independent of the mesh size h and $\boldsymbol{n}$ to stand for the unit (outer) normal vector to an edge $e \in \mathcal{E}_h$ or ∂K for $K \in \mathcal{T}_h$.

2.2. **Finite element spaces.** Let k be a non-negative integer. We define the local approximation spaces on $K \in \mathcal{T}_h$ as

$$\boldsymbol{V}(K) = P_k(K)^d, \quad W(K) = P_{k+1}(K),$$

where $P_m(K)$ stands for the space of polynomials of degree m. We introduce an approximation space for $\boldsymbol{q}|_e \cdot \boldsymbol{n}$ on $e \in \mathcal{E}_h$,

$$\boldsymbol{N}(e) = P_k(e),$$

and assume that $(I - \boldsymbol{n} \otimes \boldsymbol{n})\boldsymbol{r} = \boldsymbol{0}$ for $\boldsymbol{r} \in \boldsymbol{N}(e)$. The global finite element spaces are defined by

$$\begin{aligned}
\boldsymbol{V}_h &:= \{\boldsymbol{v} \in L^2(\Omega)^d : \boldsymbol{v}|_K \in \boldsymbol{V}(K)\ \forall K \in \mathcal{T}_h\},\\
W_h &:= \{w \in L^2(\Omega) : w|_K \in W(K)\ \forall K \in \mathcal{T}_h\},\\
\boldsymbol{N}_h &:= \{\boldsymbol{r} \in L^2(\mathcal{E}_h)^d : \boldsymbol{r}|_e \in \boldsymbol{N}(e)\ \forall e \in \mathcal{E}_h\}.
\end{aligned}$$

Let $\boldsymbol{P_V}$, P_W, and $\boldsymbol{P_N}$ denote the L^2-projections onto $\boldsymbol{V}_h$, W_h, and $\boldsymbol{N}_h$, respectively. The following approximation properties hold for $1 \le s \le k+1$: If $u \in H^{k+2}(\Omega)$, then

$$\|\boldsymbol{q} - \boldsymbol{P_V q}\| \le Ch^s |\boldsymbol{q}|_s, \tag{2a}$$

$$\|\boldsymbol{q}\cdot\boldsymbol{n} - (\boldsymbol{P_V q})\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \le Ch^{s-1/2} |\boldsymbol{q}|_s, \tag{2b}$$

$$\|u - P_W u\| \le Ch^s |u|_s, \tag{2c}$$

$$\|u - P_W u\|_{\partial\mathcal{T}_h} \le Ch^{s+1/2} |u|_{s+1}, \tag{2d}$$

$$\|(\boldsymbol{q} - \boldsymbol{P_N q})\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \le Ch^{s+1/2} |\boldsymbol{q}|_{s+1}. \tag{2e}$$

2.3. **A flux-based HDG method.** The solution of the original HDG method, $(\boldsymbol{q}_h, u_h, \widehat{u}_h) \in \boldsymbol{V}_h \times W_h \times M_h$, is defined by

$$(\boldsymbol{q}_h, \boldsymbol{v})_{\mathcal{T}_h} - (u_h, \nabla\cdot\boldsymbol{v})_{\mathcal{T}_h} + \langle \widehat{u}_h, \boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = 0 \qquad \forall \boldsymbol{v} \in \boldsymbol{V}_h, \tag{3a}$$

$$-(\boldsymbol{q}_h, \nabla w)_{\mathcal{T}_h} + \langle \widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h} = (f, w) \qquad \forall w \in W_h, \tag{3b}$$

$$\langle \widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}, \mu\rangle_{\partial\mathcal{T}_h} = 0 \qquad \forall \mu \in M_h, \tag{3c}$$

$$\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n} := \boldsymbol{q}_h\cdot\boldsymbol{n} + \tau(u_h - \widehat{u}_h) \text{ on } \partial K \qquad \forall K \in \mathcal{T}_h, \tag{3d}$$

where M_h is an approximation space for the trace $u|_{\mathcal{E}_h}$ and τ is a stabilization parameter. Another formulation via flux hybridization is also stated in [2], which reads as follows:

Find $(\boldsymbol{q}_h, u_h, \widehat{\boldsymbol{q}}_h) \in \boldsymbol{V}_h \times W_h \times \boldsymbol{N}_h$ such that

$$(\boldsymbol{q}_h, \boldsymbol{v})_{\mathcal{T}_h} - (u_h, \nabla \cdot \boldsymbol{v})_{\mathcal{T}_h} + \langle \widehat{u}_h, \boldsymbol{v} \cdot \boldsymbol{n} \rangle_{\partial \mathcal{T}_h} = 0 \qquad \forall \boldsymbol{v} \in \boldsymbol{V}_h, \tag{4a}$$

$$-(\boldsymbol{q}_h, \nabla w)_{\mathcal{T}_h} + \langle \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}, w \rangle_{\partial \mathcal{T}_h} = (f, w) \qquad \forall w \in W_h, \tag{4b}$$

$$\langle \widehat{u}_h, \boldsymbol{r} \cdot \boldsymbol{n} \rangle_{\partial \mathcal{T}_h} = 0 \qquad \forall \boldsymbol{r} \in \boldsymbol{N}_h, \tag{4c}$$

$$\widehat{u}_h := u_h + \tau^{-1}(\boldsymbol{q}_h - \widehat{\boldsymbol{q}}_h) \cdot \boldsymbol{n} \text{ on } \partial K \qquad \forall K \in \mathcal{T}_h. \tag{4d}$$

This method is a rewrite of the original HDG method and its solution coincides with that of the original method. We can verify that by expressing the hybrid variables in terms of u_h and $\boldsymbol{q}_h$. Let K^+ and K^- be adjacent elements sharing an internal edge $e \in \mathcal{E}_h$ and let $\boldsymbol{n}^+$ and $\boldsymbol{n}^-$ denote the outer unit normal vectors to ∂K^+ and ∂K^-, respectively. For a function w, let w^+ and w^- stand for the trace of $(w|_{K^+})|_e$ and $(w|_{K^-})|_e$, respectively. In both methods, $\widehat{u}_h$ and $\widehat{\boldsymbol{q}}_h$ are single valued on element boundaries from the transmission conditions (3c) and (4c). From (3d) or (4d), it follows that

$$\begin{aligned} \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}^+ &= \boldsymbol{q}_h^+ \cdot \boldsymbol{n}^+ + \tau(u_h^+ - \widehat{u}_h), \\ \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}^- &= \boldsymbol{q}_h^- \cdot \boldsymbol{n}^- + \tau(u_h^- - \widehat{u}_h). \end{aligned}$$

Solving these equations, we have

$$\begin{aligned} \widehat{u}_h &= \frac{1}{2}(u_h^+ + u_h^-) + \frac{1}{2\tau}(\boldsymbol{q}_h^+ \cdot \boldsymbol{n}^+ + \boldsymbol{q}_h^- \cdot \boldsymbol{n}^-), \\ \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}^\pm &= \frac{1}{2}(\boldsymbol{q}_h^+ + \boldsymbol{q}_h^-) \cdot \boldsymbol{n}^\pm + \frac{\tau}{2}(u_h^+ \boldsymbol{n}^+ + u_h^- \boldsymbol{n}^-) \cdot \boldsymbol{n}^\pm. \end{aligned}$$

Therefore, we see that the equations (3a)-(3b) and (4a)-(4b) give the same solution u_h and $\boldsymbol{q}_h$. However, note that the procedures of the static condensation are different and the local solvability of the flux-based method is not obvious, which we will prove later.

We consider the limiting case of $\tau \to +\infty$ in (4). In this case, (4d) is naturally interpreted as $\widehat{u}_h = u_h$, which leads to the following scheme: Find $(\boldsymbol{q}_h, u_h, \widehat{\boldsymbol{q}}_h) \in \boldsymbol{V}_h \times W_h \times \boldsymbol{N}_h$ such that

$$(\boldsymbol{q}_h, \boldsymbol{v})_{\mathcal{T}_h} - (u_h, \nabla \cdot \boldsymbol{v})_{\mathcal{T}_h} + \langle u_h, \boldsymbol{v} \cdot \boldsymbol{n} \rangle_{\partial \mathcal{T}_h} = 0 \qquad \forall \boldsymbol{v} \in \boldsymbol{V}_h, \tag{5a}$$

$$-(\boldsymbol{q}_h, \nabla w)_{\mathcal{T}_h} + \langle \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}, w \rangle_{\partial \mathcal{T}_h} = (f, w) \qquad \forall w \in W_h, \tag{5b}$$

$$\langle u_h, \boldsymbol{r} \cdot \boldsymbol{n} \rangle_{\partial \mathcal{T}_h} = 0 \qquad \forall \boldsymbol{r} \in \boldsymbol{N}_h. \tag{5c}$$

We remark that the above method is not always well-posed. Assume that $f \equiv 0$. By taking $\boldsymbol{v} = \boldsymbol{q}_h$, $w = u_h$, and $\boldsymbol{r} = \widehat{\boldsymbol{q}}_h$ in (5), we have $\boldsymbol{q}_h = \boldsymbol{0}$. From (5a) with $\boldsymbol{q}_h = \boldsymbol{0}$ and (5c), it follows that $u_h = 0$. However, $\widehat{\boldsymbol{q}}_h$ still remains unknown, which depends on if the following equation implies $\widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n} = 0$:

$$\langle \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}, w \rangle_{\partial \mathcal{T}_h} = 0 \qquad \forall w \in W_h. \tag{6}$$

Indeed, when $\boldsymbol{N}_h$ and W_h are piecewise constant spaces, it is easy to see that there exists $\widehat{\boldsymbol{q}}_h \in \boldsymbol{N}_h$ satisfying (6) and $\widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n} \neq 0$. For this reason, we need the following inf-sup

condition for the well-posedness: There exists a constant C independent of h such that, for all $\boldsymbol{r} \in \boldsymbol{N}_h$,

$$\|h^{1/2}\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \le C \sum_{K\in\mathcal{T}_h} \sup_{w\in W(K)} \frac{\langle \boldsymbol{r}\cdot\boldsymbol{n}, w\rangle_{\partial K}}{\|\nabla w\|_{L^2(K)} + \|h^{-1/2}\boldsymbol{P}_{\boldsymbol{N}} w\|_{L^2(\partial K)}}, \tag{7}$$

where $\boldsymbol{P}_{\boldsymbol{N}} w := \boldsymbol{P}_{\boldsymbol{N}}(w|_{\partial K}\boldsymbol{n})\cdot\boldsymbol{n}$. In order to derive a priori estimates, we also use the transposed version of the inf-sup condition: There exists a constant C such that

$$\|h^{-1/2}\boldsymbol{P}_{\boldsymbol{N}} w\|_{\partial\mathcal{T}_h} \le C \sum_{K\in\mathcal{T}_h} \sup_{\boldsymbol{r}\in\boldsymbol{N}(\partial K)} \frac{\langle \boldsymbol{r}\cdot\boldsymbol{n}, w\rangle_{\partial K}}{\|h^{1/2}\boldsymbol{r}\cdot\boldsymbol{n}\|_{L^2(\partial K)}} \qquad \forall w \in W_h. \tag{8}$$

2.4. **Local solvability.** We here verify the local solvability of the flux-based methods (4) and (5).

We first consider the local problem of (4). Let $\widehat{\boldsymbol{q}}_{\partial K}$ denote the restriction of $\widehat{\boldsymbol{q}}_h$ to ∂K and let $[\![w]\!]$ denote the jump of a function w. We define $\|\mu\|_{\mathcal{E}_h} = \langle \mu, \mu\rangle_{\mathcal{E}_h}^{1/2}$ and $\|\boldsymbol{r}\|_{\mathcal{E}_h} = \langle \boldsymbol{r}, \boldsymbol{r}\rangle_{\mathcal{E}_h}^{1/2}$. We introduce the mean-zero subspace of $W(K)$,

$$W_0(K) := \{w \in W(K) : (w, 1)_K = 0\}.$$

The local problem reads: Find $(\boldsymbol{q}_K, u_{K0}) \in \boldsymbol{V}(K) \times W_0(K)$ such that

$$(\boldsymbol{q}_K, \boldsymbol{v})_K + \langle \tau^{-1}\boldsymbol{q}_K\cdot\boldsymbol{n}, \boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial K} + (\nabla u_{K0}, \boldsymbol{v})_K = \langle \tau^{-1}\widehat{\boldsymbol{q}}_{\partial K}\cdot\boldsymbol{n}, \boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial K} \qquad \forall \boldsymbol{v} \in \boldsymbol{V}(K), \tag{9a}$$

$$-(\boldsymbol{q}_K, \nabla w_0)_K = (f, w_0)_K - \langle \widehat{\boldsymbol{q}}_{\partial K}\cdot\boldsymbol{n}, w_0\rangle_{\partial K} \quad \forall w_0 \in W_0(K). \tag{9b}$$

To verify the well-posedness of the local problem, we let $f \equiv 0$ and $\widehat{\boldsymbol{q}}_{\partial K}\cdot\boldsymbol{n} = 0$. Taking $\boldsymbol{v} = \boldsymbol{q}_K$ in (9a) and $w_0 = u_{K0}$ in (9b), we have $\boldsymbol{q}_K = \boldsymbol{0}$. Since we can $\boldsymbol{v} = \nabla u_{K0}$, we get $\nabla u_{K0} = 0$, which implies $u_{K0} = 0$. Therefore, $\boldsymbol{q}_K$ and u_{K0} are uniquely determined if $\widehat{\boldsymbol{q}}_{\partial K}$ is given.

However, the piecewise constant part of u_h, denoted by $\overline{u}_h$, remains unknown. Eliminating $\boldsymbol{q}_K$ and u_{K0} in element-by-element fashion by static condensation, we obtain the global equations for $(\widehat{\boldsymbol{q}}_h, \overline{u}_h) \in \boldsymbol{N}_h \times P_0(\mathcal{T}_h)$

$$\langle \overline{u}_h - \tau^{-1}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = -\langle u_{h0} - \tau^{-1}\boldsymbol{q}_h\cdot\boldsymbol{n}, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} =: F_1(\boldsymbol{r}) \quad \forall \boldsymbol{r} \in \boldsymbol{N}_h, \tag{10a}$$

$$\langle \widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}, \overline{w}\rangle_{\partial\mathcal{T}_h} = (f, \overline{w}) \qquad \forall \overline{w} \in P_0(\mathcal{T}_h), \tag{10b}$$

where $u_{h0} := u_h - \overline{u}_h$ and we note that u_{h0} has been determined by (9). We will show that the global problem is well posed. To this end, we first prove the following inf-sup condition.

Theorem 2.1. There exists a positive constant C such that

$$C\|[\![\overline{w}]\!]\|_{\mathcal{E}_h} \le \sup_{\boldsymbol{r}\in\boldsymbol{N}_h} \frac{\langle \boldsymbol{r}\cdot\boldsymbol{n}, \overline{w}\rangle_{\partial\mathcal{T}_h}}{\|\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}} \qquad \forall \overline{w} \in P_0(\mathcal{E}_h).$$

Proof. For $\overline{w} \in P_0(\mathcal{T}_h)$ with $\overline{w} \ne 0$, we define $\boldsymbol{r} = [\![\overline{w}]\!]$. Then, it follows that

$$\langle \boldsymbol{r}\cdot\boldsymbol{n}, \overline{w}\rangle_{\partial\mathcal{T}_h} = \langle \boldsymbol{r}, [\![\overline{w}]\!]\rangle_{\mathcal{E}_h} = \|[\![\overline{w}]\!]\|_{\mathcal{E}_h}^2 = \|\boldsymbol{r}\|_{\mathcal{E}_h}\|[\![\overline{w}]\!]\|_{\mathcal{E}_h}.$$

Since $\|\boldsymbol{r}\|_{\mathcal{E}_h} \le \|\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \le \sqrt{2}\|\boldsymbol{r}\|_{\mathcal{E}_h}$ for $\boldsymbol{r}\in\boldsymbol{N}_h$, we have

$$\|[\![\overline{w}]\!]\|_{\mathcal{E}_h} \le \frac{\langle \boldsymbol{r}\cdot\boldsymbol{n}, \overline{w}\rangle_{\partial\mathcal{T}_h}}{\|\boldsymbol{r}\|_{\mathcal{E}_h}} \le \sqrt{2}\cdot\frac{\langle \boldsymbol{r}\cdot\boldsymbol{n}, \overline{w}\rangle_{\partial\mathcal{T}_h}}{\|\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}},$$

which completes the proof. □

We now prove an a priori estimate for the global problem (10) using the inf-sup condition in Theorem 2.1, which ensures that the problem admits a unique solution.

Theorem 2.2. There exists a positive constant C independent of h such that

$$\|\tau^{-1/2}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} + \|[\![\overline{u}_h]\!]\|_{\mathcal{E}_h} \le C\left(\|u_{h0}\|_{\partial\mathcal{T}_h} + \|\boldsymbol{q}_h\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} + \|f\|\right).$$

Proof. Taking $\boldsymbol{r} = \widehat{\boldsymbol{q}}_h$ in (10a) and $\overline{w} = \overline{u}_h$ in (10b), we have

$$\begin{aligned}\langle \overline{u}_h, \widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - \|\tau^{-1/2}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|^2_{\partial\mathcal{T}_h} &= F_1(\widehat{\boldsymbol{q}}_h),\\ \langle \widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}, \overline{u}_h\rangle_{\partial\mathcal{T}_h} &= (f,\overline{u}_h).\end{aligned}$$

It then follows that

$$\|\tau^{-1/2}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|^2_{\partial\mathcal{T}_h} = (f,\overline{u}_h) - F_1(\widehat{\boldsymbol{q}}_h) \le \|f\|\|\overline{u}_h\| + \|F_1\|\|\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h},$$

where

$$\|F_1\| := \sup_{\boldsymbol{r}\in\boldsymbol{N}_h}\frac{F(\boldsymbol{r})}{\|\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}}.$$

Using Young's inequality, we deduce

$$\|\tau^{-1/2}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|^2_{\partial\mathcal{T}_h} \le C\left(\|F_1\|^2 + \epsilon^{-2}\|f\|^2\right) + \epsilon^2\|\overline{u}_h\|^2_{\partial\mathcal{T}_h} \tag{11}$$

for any $\epsilon > 0$. By Theorem 2.1, (10a) and (11), we have

$$\begin{aligned}\|[\![\overline{u}_h]\!]\|_{\mathcal{E}_h} &\le C\sup_{\boldsymbol{r}\in\boldsymbol{N}_h}\frac{\langle \boldsymbol{r}\cdot\boldsymbol{n}, \overline{u}_h\rangle_{\partial\mathcal{T}_h}}{\|\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}}\\ &\le C\sup_{\boldsymbol{r}\in\boldsymbol{N}_h}\frac{F_1(\boldsymbol{r}) + \langle \tau^{-1}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}}{\|\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}}\\ &\le C(\|F_1\| + \|\tau^{-1}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h})\\ &\le C\left(\|F_1\| + \epsilon^{-1}\|f\| + \epsilon\|\overline{u}_h\|_{\partial\mathcal{T}_h}\right).\end{aligned}$$

Since both $\|[\![\cdot]\!]\|_{\mathcal{E}_h}$ and $\|\cdot\|_{\partial\mathcal{T}_h}$ are norms on $P_0(\mathcal{T}_h)$, they are equivalent to each other and $\|\overline{u}_h\|_{\partial\mathcal{T}_h}$ is bounded by $C\|[\![\overline{u}_h]\!]\|_{\mathcal{E}_h}$. Choosing ϵ sufficiently small, we get

$$\|[\![\overline{u}_h]\!]\|_{\mathcal{E}_h} \le C\left(\|F_1\| + \|f\|\right).$$

From this and (11), it follows that

$$\|\tau^{-1/2}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \le C\left(\|F_1\| + \|f\|\right).$$

Since we can bound as $\|F_1\| \le C(\|u_{h0}\|_{\partial\mathcal{T}_h} + \|\boldsymbol{q}_h\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h})$, the proof is complete. □

Next, we show that the local problem of the proposed method is well-posed. Let us define $\boldsymbol{R}_0(\partial K) = \{w|_{\partial K}\boldsymbol{n} : w \in P_0(K)\}$, whose dimension is one. The local problem of (5) is as follows: Find $(\boldsymbol{q}_K, u_{K0}, \overline{u}_K) \in \boldsymbol{V}(K) \times W_0(K) \times P_0(K)$ such that

$$(\boldsymbol{q}_K, \boldsymbol{v})_K + (\nabla u_{K0}, \boldsymbol{v})_K = 0 \qquad \forall \boldsymbol{v} \in \boldsymbol{V}(K), \tag{12a}$$

$$-(\boldsymbol{q}_K, \nabla w_0)_K = (f, w_0)_K - \langle \widehat{\boldsymbol{q}}_{\partial K} \cdot \boldsymbol{n}, w_0 \rangle_{\partial K} \qquad \forall w_0 \in W_0(K), \tag{12b}$$

$$\langle \overline{u}_K + u_{K0}, \boldsymbol{r}_0 \cdot \boldsymbol{n} \rangle_{\partial K} = 0 \qquad \forall \boldsymbol{r}_0 \in \boldsymbol{R}_0(\partial K). \tag{12c}$$

The well-posedness is verified by setting all terms on the right-hand side to zero and a straightforward computation.

3. Error Analysis

3.1. **A priori estimate.**

We consider the proposed method in general form

$$(\boldsymbol{q}_h, \boldsymbol{v})_{\mathcal{T}_h} - (u_h, \nabla \cdot \boldsymbol{v})_{\mathcal{T}_h} + \langle u_h, \boldsymbol{v} \cdot \boldsymbol{n} \rangle_{\partial \mathcal{T}_h} = F_1(\boldsymbol{v}) \qquad \forall \boldsymbol{v} \in \boldsymbol{V}_h, \tag{13a}$$

$$-(\boldsymbol{q}_h, \nabla w)_{\mathcal{T}_h} + \langle \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}, w \rangle_{\partial \mathcal{T}_h} = F_2(w) \qquad \forall w \in W_h, \tag{13b}$$

$$\langle u_h, \boldsymbol{r} \cdot \boldsymbol{n} \rangle_{\partial \mathcal{T}_h} = F_3(\boldsymbol{r}) \qquad \forall \boldsymbol{r} \in \boldsymbol{N}_h, \tag{13c}$$

where $F_1 : \boldsymbol{V}_h \to \mathbb{R}$, $F_2 : W_h \to \mathbb{R}$, and $F_3 : \boldsymbol{N}_h \to \mathbb{R}$ are linear functionals and their norms are defined by

$$\|F_1\| = \sup_{\boldsymbol{v} \in \boldsymbol{V}_h} \frac{F_1(\boldsymbol{v})}{\|\boldsymbol{v}\|},$$

$$\|F_2\| = \sum_{K \in \mathcal{T}_h} \sup_{w \in W(K)} \frac{F_2(w)}{\|\nabla w\|_{L^2(K)} + \|h^{-1/2} \boldsymbol{P}_{\boldsymbol{N}} w\|_{L^2(\partial K)}},$$

$$\|F_3\| = \sup_{\boldsymbol{r} \in \boldsymbol{N}_h} \frac{F_3(\boldsymbol{r})}{\|h^{1/2} \boldsymbol{r} \cdot \boldsymbol{n}\|_{\partial \mathcal{T}_h}}.$$

We first establish a priori estimate for the problem.

Theorem 3.1. Let $(\boldsymbol{q}_h, u_h, \widehat{\boldsymbol{q}}_h) \in \boldsymbol{V}_h \times W_h \times \boldsymbol{N}_h$ be a solution of (13). Then there exists a constant C such that

$$\|\boldsymbol{q}_h\| + \|h^{1/2} \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}\|_{\partial \mathcal{T}_h} + \|\nabla u_h\|_{\mathcal{T}_h} + \|h^{-1/2} \boldsymbol{P}_{\boldsymbol{N}} u_h\|_{\partial \mathcal{T}_h} \leq C \left(\|F_1\| + \|F_2\| + \|F_3\| \right).$$

Proof. From the inf-sup condition (7) and (13b), it follows that

$$\begin{aligned} \|h^{1/2} \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}\|_{\partial \mathcal{T}_h} &\leq C \sum_{K \in \mathcal{T}_h} \sup_{w \in W(K)} \frac{\langle \widehat{\boldsymbol{q}}_h \cdot \boldsymbol{n}, w \rangle_{\partial K}}{\|\nabla w\|_{L^2(K)} + \|h^{-1/2} \boldsymbol{P}_{\boldsymbol{N}} w\|_{L^2(\partial K)}} \\ &= C \sum_{K \in \mathcal{T}_h} \sup_{w \in W(K)} \frac{F_2(w) + (\boldsymbol{q}_h, \nabla w)_K}{\|\nabla w\|_{L^2(K)} + \|h^{-1/2} \boldsymbol{P}_{\boldsymbol{N}} w\|_{L^2(\partial K)}} \\ &\leq C \left(\|F_2\| + \|\boldsymbol{q}_h\| \right). \end{aligned} \tag{14}$$

Integrating by parts in (13a), we have

$$(\boldsymbol{q}_h, \boldsymbol{v})_{\mathcal{T}_h} + (\nabla u_h, \boldsymbol{v})_{\mathcal{T}_h} = F_1(\boldsymbol{v}). \tag{15}$$

Substituting $\boldsymbol{v} = \nabla u_h$ in the above equation, we get

$$\|\nabla u_h\|_{\mathcal{T}_h} \leq C\left(\|\boldsymbol{q}_h\| + \|F_1\|\right). \tag{16}$$

Taking $\boldsymbol{v} = \boldsymbol{q}_h$ in (15), $\boldsymbol{r} = \widehat{\boldsymbol{q}}_h$ in (13c), and $w = u_h$ in (13b), we have

$$\begin{aligned}\|\boldsymbol{q}_h\|^2 &= F_1(\boldsymbol{q}_h) + F_2(u_h) - F_3(\widehat{\boldsymbol{q}}_h)\\ &\leq \|F_1\|\|\boldsymbol{q}_h\| + C\|F_2\| \sum_{K\in\mathcal{T}_h}\left(\|\nabla u_h\|_{L^2(K)} + \|h^{-1/2}\boldsymbol{P}_{\boldsymbol{N}}u_h\|_{\partial K}\right) + \|F_3\|\|h^{1/2}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}.\end{aligned}$$

By (8), we have

$$\begin{aligned}\|h^{-1/2}\boldsymbol{P}_{\boldsymbol{N}}u_h\|_{\partial\mathcal{T}_h} &\leq C\sum_{K\in\mathcal{T}_h}\sup_{\boldsymbol{r}\in\boldsymbol{N}_h}\frac{\langle\boldsymbol{r}\cdot\boldsymbol{n}, \boldsymbol{P}_{\boldsymbol{N}}u_h\rangle_{\partial\mathcal{T}_h}}{\|h^{1/2}\boldsymbol{r}\cdot\boldsymbol{n}\|}\\ &= C\sum_{K\in\mathcal{T}_h}\sup_{\boldsymbol{r}\in\boldsymbol{N}_h}\frac{\langle\boldsymbol{r}\cdot\boldsymbol{n}, u_h\rangle_{\partial\mathcal{T}_h}}{\|h^{1/2}\boldsymbol{r}\cdot\boldsymbol{n}\|}\\ &\leq C\sum_{K\in\mathcal{T}_h}\sup_{\boldsymbol{r}\in\boldsymbol{N}_h}\frac{F_3(\boldsymbol{r})}{\|h^{1/2}\boldsymbol{r}\cdot\boldsymbol{n}\|}\\ &\leq C\|F_3\|.\end{aligned}$$

Combining this with (16), we get

$$\sum_{K\in\mathcal{T}_h}\left(\|\nabla u_h\|_{L^2(K)} + \|h^{-1/2}\boldsymbol{P}_{\boldsymbol{N}}u_h\|_{\partial K}\right) \leq C\left(\|\boldsymbol{q}_h\| + \|F_3\|\right).$$

Thus we estimate as

$$\|\boldsymbol{q}_h\|^2 \leq \|F_1\|\|\boldsymbol{q}_h\| + C\|F_2\|\left(\|\boldsymbol{q}_h\| + \|F_1\|\right) + C\|F_3\|(\|F_2\| + \|\boldsymbol{q}_h\|).$$

Using Young's inequality, we obtain

$$\|\boldsymbol{q}_h\|^2 \leq C\left(\|F_1\|^2 + \|F_2\|^2 + \|F_3\|^2\right).$$

From this and (14), it follows that

$$\|h^{1/2}\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \leq C\left(\|F_2\| + \|\boldsymbol{q}_h\|\right) \leq C(\|F_1\| + \|F_2\| + \|F_3\|),$$

which completes the proof. □

If $f \equiv 0$, it follows from Theorem 3.1 that $\boldsymbol{q}_h = \boldsymbol{0}$, u_h is constant on each element, $\widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n} = 0$, and $\boldsymbol{P}_{\boldsymbol{N}}u_h = 0$ on element boundaries. Thus, we have verified the existence and uniqueness of our method.

3.2. **Optimal convergence of $\boldsymbol{q}_h$.** The projections of errors are defined as

$$\boldsymbol{e}_{\boldsymbol{q}} = \boldsymbol{P}_{\boldsymbol{V}}\boldsymbol{q} - \boldsymbol{q}_h, \quad e_u = P_W u - u_h, \quad \boldsymbol{e}_{\widehat{\boldsymbol{q}}}\cdot\boldsymbol{n} = \boldsymbol{P}_{\boldsymbol{N}}\boldsymbol{q}\cdot\boldsymbol{n} - \widehat{\boldsymbol{q}}_h\cdot\boldsymbol{n}.$$

Theorem 3.2. If $u \in H^{k+2}(\Omega)$, then we have

$$\|\boldsymbol{e}_{\boldsymbol{q}}\| + \|h^{1/2}\boldsymbol{e}_{\widehat{\boldsymbol{q}}}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \leq Ch^{k+1}|u|_{k+2}.$$

Proof. The problem (1) is rewritten into

$$\begin{aligned}(\boldsymbol{q},\boldsymbol{v})_{\mathcal{T}_h} - (u,\nabla\cdot\boldsymbol{v})_{\mathcal{T}_h} + \langle u,\boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} &= 0 && \forall \boldsymbol{v}\in\boldsymbol{V}_h,\\ -(\boldsymbol{q},\nabla w)_{\mathcal{T}_h} + \langle \boldsymbol{q}\cdot\boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h} &= (f,w) && \forall w\in W_h,\\ \langle u,\boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} &= 0 && \forall \boldsymbol{r}\in\boldsymbol{N}_h.\end{aligned}$$

By the property of the L^2-projections, the above equations become

$$(\boldsymbol{P_V q},\boldsymbol{v})_{\mathcal{T}_h} - (P_W u,\nabla\cdot\boldsymbol{v})_{\mathcal{T}_h} + \langle P_W u,\boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = G_1(\boldsymbol{v}) \qquad \forall \boldsymbol{v}\in\boldsymbol{V}_h, \tag{17a}$$

$$-(\boldsymbol{P_V q},\nabla w)_{\mathcal{T}_h} + \langle \boldsymbol{P_N q}\cdot\boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h} = (f,w) + G_2(w) \qquad \forall w\in W_h, \tag{17b}$$

$$\langle P_W u,\boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = G_3(\boldsymbol{r}) \qquad \forall \boldsymbol{r}\in\boldsymbol{N}_h, \tag{17c}$$

where we have integrated by parts in the first equation and

$$\begin{aligned}G_1(\boldsymbol{v}) &:= -\langle u - P_W u,\boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h},\\ G_2(w) &:= -\langle \boldsymbol{q}\cdot\boldsymbol{n} - \boldsymbol{P_N q}\cdot\boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h},\\ G_3(\boldsymbol{r}) &:= -\langle u - P_W u,\boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}.\end{aligned}$$

The norms of G_1 and G_3 are bounded as

$$\|G_1\| = \sup_{\boldsymbol{v}\in\boldsymbol{V}_h}\frac{G_1(\boldsymbol{v})}{\|\boldsymbol{v}\|} \le \|u - P_W u\|_{\partial\mathcal{T}_h}\cdot Ch^{-1/2} \le Ch^{k+1}|u|_{k+2},$$

$$\|G_3\| = \sup_{\boldsymbol{r}\in\boldsymbol{N}_h}\frac{G_3(\boldsymbol{r})}{\|h^{1/2}\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}} \le \|u - P_W u\|_{\partial\mathcal{T}_h}\cdot h^{-1/2} \le Ch^{k+1}|u|_{k+2}.$$

Using [6, Lemma 3], we can estimate

$$\begin{aligned}|G_2(w)| &\le |\langle(\boldsymbol{q} - \boldsymbol{P_N q})\cdot\boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h}|\\ &= \|(\boldsymbol{q} - \boldsymbol{P_N q})\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}\cdot Ch^{1/2}\|\nabla w\|_{\mathcal{T}_h}.\end{aligned}$$

The norm of G_2 is bounded as

$$\|G_2\| = \sum_{K\in\mathcal{T}_h}\sup_{w\in W(K)}\frac{G_2(w)}{\|\nabla w\|_{L^2(K)} + \|h^{-1/2}\boldsymbol{P_N} w\|_{L^2(\partial K)}} \le Ch^{k+1}|\boldsymbol{q}|_{k+1}.$$

Subtracting (5) from (17), we obtain the error equations

$$(\boldsymbol{e_q},\boldsymbol{v})_{\mathcal{T}_h} - (e_u,\nabla\cdot\boldsymbol{v})_{\mathcal{T}_h} + \langle e_u,\boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = G_1(\boldsymbol{v}) \qquad \forall \boldsymbol{v}\in\boldsymbol{V}_h, \tag{18a}$$

$$-(\boldsymbol{e_q},\nabla w)_{\mathcal{T}_h} + \langle \boldsymbol{e_{\widehat{q}}}\cdot\boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h} = G_2(w) \qquad \forall w\in W_h, \tag{18b}$$

$$\langle e_u,\boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = G_3(\boldsymbol{r}) \qquad \forall \boldsymbol{r}\in\boldsymbol{N}_h. \tag{18c}$$

Applying Theorem 3.1 to the error equations leads to

$$\|\boldsymbol{e_q}\| + \|h^{1/2}\boldsymbol{e_{\widehat{q}}}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \le C\left(\|G_1\| + \|G_2\| + \|G_3\|\right) \le Ch^{k+1}|u|_{k+2}.$$

□

From Theorem 3.1, it also follows that

$$\|\nabla e_u\|_{\mathcal{T}_h} + \|h^{-1/2}\boldsymbol{P_N} e_u\|_{\partial\mathcal{T}_h} \le Ch^{k+1}|u|_{k+2}. \tag{19}$$

3.3. **L^2-error estimate of u_h.** We consider the following adjoint problem: Find $(\boldsymbol{\theta}, \xi) \in \boldsymbol{H}^1(\Omega) \times (H^2(\Omega) \cap H_0^1(\Omega))$ such that

$$\begin{aligned} \boldsymbol{\theta} + \nabla \xi &= \boldsymbol{0} && \text{in } \Omega, \\ \nabla \cdot \boldsymbol{\theta} &= e_u && \text{in } \Omega, \\ \xi &= 0 && \text{on } \partial\Omega. \end{aligned}$$

It is well known that the elliptic regularity holds:

$$\|\boldsymbol{\theta}\|_1 + \|\xi\|_2 \le C\|e_u\|.$$

We provide an L^2-error estimate of u_h by the Aubin–Nitsche technique.

Theorem 3.3. If $k \ge 1$ and $u \in H^{k+2}(\Omega)$, then there exists a constant C such that

$$\|e_u\| \le Ch^{k+2}|u|_{k+2}.$$

Proof. Since (17) holds for the adjoint problem, we have

$$(\boldsymbol{P_V\theta}, \boldsymbol{v})_{\mathcal{T}_h} - (P_W\xi, \nabla\cdot\boldsymbol{v})_{\mathcal{T}_h} + \langle P_W\xi, \boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = G_1'(\boldsymbol{v}) \qquad \forall \boldsymbol{v} \in \boldsymbol{V}_h, \tag{21a}$$

$$-(\boldsymbol{P_V\theta}, \nabla w)_{\mathcal{T}_h} + \langle \boldsymbol{P_N\theta}\cdot\boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h} = (e_u, w) + G_2'(w) \qquad \forall w \in W_h, \tag{21b}$$

$$\langle P_W\xi, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = G_3'(\boldsymbol{r}) \qquad \forall \boldsymbol{r} \in \boldsymbol{N}_h, \tag{21c}$$

where

$$\begin{aligned} G_1'(\boldsymbol{v}) &:= -\langle \xi - P_W\xi, \boldsymbol{v}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}, \\ G_2'(w) &:= -\langle (\boldsymbol{\theta} - \boldsymbol{P_N\theta})\cdot\boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h}, \\ G_3'(\boldsymbol{r}) &:= -\langle \xi - P_W\xi, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}. \end{aligned}$$

Taking $\boldsymbol{v} = -\boldsymbol{e_q}$ in (21a), $w = e_u$ in (21b), and $\boldsymbol{r} = \boldsymbol{e_{\widehat{q}}}$ in (21c), we have

$$\begin{aligned} &-(\boldsymbol{P_V\theta}, \boldsymbol{e_q})_{\mathcal{T}_h} - (\nabla P_W\xi, \boldsymbol{e_q})_{\mathcal{T}_h} - (\boldsymbol{P_V\theta}, \nabla e_u)_{\mathcal{T}_h} + \langle \boldsymbol{P_N\theta}\cdot\boldsymbol{n}, e_u\rangle_{\partial\mathcal{T}_h} + \langle P_W\xi, \boldsymbol{e_{\widehat{q}}}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} \\ &\qquad = \|e_u\|^2 - G_1'(\boldsymbol{e_q}) + G_2'(e_u) + G_3'(\boldsymbol{e_{\widehat{q}}}). \end{aligned} \tag{22}$$

Choosing $\boldsymbol{v} = -\boldsymbol{P_V\theta}$ in (18a), $w = P_W\xi$ in (18b), and $\boldsymbol{r} = \boldsymbol{P_N\theta}$ in (18c), we have

$$\begin{aligned} &-(\boldsymbol{e_q}, \boldsymbol{P_V\theta})_{\mathcal{T}_h} - (\nabla e_u, \boldsymbol{P_V\theta})_{\mathcal{T}_h} - (\boldsymbol{e_q}, \nabla P_W\xi)_{\mathcal{T}_h} + \langle \boldsymbol{e_{\widehat{q}}}\cdot\boldsymbol{n}, P_W\xi\rangle_{\partial\mathcal{T}_h} + \langle e_u, \boldsymbol{P_N\theta}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} \\ &\qquad = -G_1(\boldsymbol{P_V\theta}) + G_2(P_W\xi) + G_3(\boldsymbol{P_N\theta}). \end{aligned} \tag{23}$$

Subtracting (23) from (22) yields

$$\|e_u\|^2 = -G_1(\boldsymbol{P_V\theta}) + G_2(P_W\xi) + G_3(\boldsymbol{P_N\theta}) - G_1'(\boldsymbol{e_q}) + G_2'(e_u) + G_3'(\boldsymbol{e_{\widehat{q}}}).$$

We will bound the terms on the right-hand side. The first and third terms are bounded as

$$\begin{aligned} |-G_1(\boldsymbol{P_V\theta}) + G_3(\boldsymbol{P_N\theta})| &= |\langle u - P_Wu, (\boldsymbol{P_V\theta} - \boldsymbol{P_N\theta})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}| \\ &\le \|u - P_Wu\|_{\partial\mathcal{T}_h}(\|(\boldsymbol{P_V\theta} - \boldsymbol{\theta})\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} + \|(\boldsymbol{\theta} - \boldsymbol{P_N\theta})\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}) \\ &\le Ch^{k+3/2}|u|_{k+2}\cdot Ch^{1/2}|\boldsymbol{\theta}|_1 \\ &= Ch^{k+2}|u|_{k+2}|\boldsymbol{\theta}|_1. \end{aligned}$$

Let P_1 denote the L^2-projection from $L^2(\Omega)$ onto $P_1(\mathcal{T}_h)$. Note that $\langle(\boldsymbol{q}-\boldsymbol{P_N q})\cdot\boldsymbol{n}, P_1\xi\rangle_{\partial\mathcal{T}_h} = 0$ since we assume $k \geq 1$. We have

$$
\begin{aligned}
|G_2(P_W\xi)| &= |\langle(\boldsymbol{q}-\boldsymbol{P_N q})\cdot\boldsymbol{n}, P_W\xi - P_1\xi\rangle_{\partial\mathcal{T}_h}| \\
&\leq \|(\boldsymbol{q}-\boldsymbol{P_N q})\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}(\|P_W\xi-\xi\|_{\partial\mathcal{T}_h} + \|\xi - P_1\xi\|_{\partial\mathcal{T}_h}) \\
&\leq Ch^{k+1/2}|\boldsymbol{q}|_{k+1}\cdot Ch^{3/2}|\xi|_2 \\
&= Ch^{k+2}|\boldsymbol{q}|_{k+1}|\xi|_2.
\end{aligned} \tag{24}
$$

The rest terms are bounded as follows:

$$
\begin{aligned}
|G_1'(\boldsymbol{e_q})| &\leq Ch|\xi|_2\|\boldsymbol{e_q}\|, \\
|G_2'(e_u)| &\leq Ch|\boldsymbol{\theta}|_1\|\nabla e_u\|_{\mathcal{T}_h} \qquad \text{(by [6, Lemma 3])} \\
&\leq Ch|\boldsymbol{\theta}|_1\left(\|\boldsymbol{e_q}\| + h^{k+1}|u|_{k+2}\right), \qquad \text{(by (19))} \\
|G_3'(\boldsymbol{e}_{\widehat{\boldsymbol{q}}})| &\leq Ch|\xi|_2\|h^{1/2}\boldsymbol{e}_{\widehat{\boldsymbol{q}}}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h}.
\end{aligned}
$$

Thus we deduce

$$
\|e_u\| \leq C\left(h\|\boldsymbol{e_q}\| + h\|h^{1/2}\boldsymbol{e}_{\widehat{\boldsymbol{q}}}\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} + h^{k+2}|u|_{k+2}\right) \leq Ch^{k+2}|u|_{k+2}.
$$

□

Remark 3.4. Theorem 3.3 also holds for $k = 0$. When $k = 0$, we can use the Crouzeix–Raviart interpolation I_{CR} instead of P_W. Then, the right-hand sides of (18) and (21) are changed as $G_1(\boldsymbol{v}) = -(\nabla(u - I_{CR}u), \boldsymbol{v})_{\mathcal{T}_h}$, $G_3(\boldsymbol{r}) = 0$, $G_1'(\boldsymbol{v}) = -(\nabla(\xi - I_{CR}\xi), \boldsymbol{v})_{\mathcal{T}_h}$, $G_3'(\boldsymbol{r}) = 0$. It is clear that they are bounded by the Schwarz inequality and the interpolation error estimate, and we do not need to use the L^2-projection in (24) since $G_2(I_{CR}\xi) = \langle(\boldsymbol{q} - \boldsymbol{P_N q})\cdot\boldsymbol{n}, I_{CR}\xi\rangle_{\partial\mathcal{T}_h} = 0$.

4. Proof of the inf-sup condition for triangular elements

We show that the inf-sup conditions (7) and (8) are satisfied for the triangular P_k-element in the two-dimensional case.

Let T_1 and T_2 be the reference triangles whose vertices are $\{(0,0),(0,1),(1,1)\}$ and $\{(1,0),(0,0),(0,1)\}$, respectively. Let $\{e_1, e_2, e_3\}$ and $\{e_3, e_4, e_5\}$ denote be the edges of T_1 and T_2, respectively, see Figure 1. For $1 \leq i \leq 5$, we define F_i by the linear transforms from e_i to $[-1, 1]$ such that

$$
\begin{aligned}
F_1((0,0)) &= -1, \quad F_1((1,0)) = 1, \\
F_2((1,0)) &= -1, \quad F_2((1,1)) = 1, \\
F_3((1,1)) &= -1, \quad F_3((0,0)) = 1, \\
F_4((0,0)) &= -1, \quad F_4((0,1)) = 1, \\
F_5((0,1)) &= -1, \quad F_5((1,1)) = 1.
\end{aligned}
$$

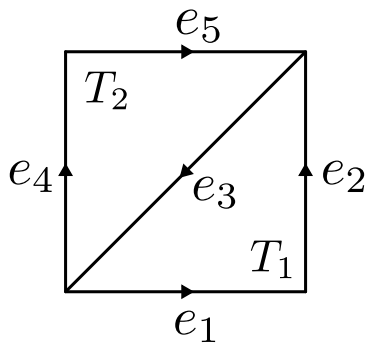


FIGURE 1. The reference triangles T_1 and T_2 and the orientation of the edges $e_1, \dots, e_5$

Let $\chi^{(i)}$ denote the characteristic function of e_i and let φ_p denote the Legendre polynomial of degree p on $[-1, 1]$. We define the normalized Legendre polynomial on e_i by

$$\varphi_p^{(i)} = \frac{(\varphi_p \circ F_i)\chi^{(i)}}{\left(\int_{e_i} (\varphi_p \circ F_i)^2 ds\right)^{1/2}}$$

for $0 \le p \le k$. It is clear that $\{\varphi_0^{(i)}, \varphi_1^{(i)}, \dots, \varphi_k^{(i)}\}$ is a basis of $\boldsymbol{N}(e_i) = P_k(e_i)$ and satisfies the orthogonality

$$\langle \varphi_p^{(i)}, \varphi_q^{(j)} \rangle_{\partial T_1 \cup \partial T_2} = \delta_{ij}\delta_{pq} \quad \text{for } 1 \le i, j, p, q \le 5.$$

Note that $\varphi_p^{(i)} = (-1)^p$ at the starting point of e_i and $\varphi_p^{(i)} = 1$ at the end point of e_i.

Lemma 4.1. Let $\boldsymbol{r} \in \bigoplus_{1 \le i \le 5} \boldsymbol{N}(e_i)$. It holds that $\boldsymbol{r} \cdot \boldsymbol{n} = 0$ on e_i $(1 \le i \le 5)$ if and only if

$$\langle \boldsymbol{r} \cdot \boldsymbol{n}, w \rangle_{\partial T_1 \cup \partial T_2} = 0 \quad \forall w \in W(T_1) \oplus W(T_2). \tag{25}$$

Proof. We prove that only when k is even, i.e., $k = 2k'$, since the proof when k is odd is similar. We show that $\boldsymbol{r} \cdot \boldsymbol{n} = 0$ follows from (25). We can write $\boldsymbol{r} \cdot \boldsymbol{n}$ as

$$\boldsymbol{r} \cdot \boldsymbol{n} = \sum_{i=1}^{5} \sum_{p=0}^{2k'} a_p^{(i)} \varphi_p^{(i)}, \quad a_p^{(i)} \in \mathbb{R}.$$

First, we show that $a_{2q}^{(i)} = 0$ for $1 \le q \le k'$ and $1 \le i \le 5$. Let us define

$$w_1 = \varphi_{2q}^{(1)} - \varphi_{2k'+1}^{(2)} + \varphi_{2k'+1}^{(3)}$$

for $1 \le q \le k'$, see also Figure 2. Since w_1 is continuous at the vertices, there exists $Ew_1 \in W(T_1) \oplus W(T_2)$ such that $Ew_1 = w_1$ on element boundaries. For simplicity,

we use the same symbol w_1 to denote Ew_1. Choosing $w = w_1$ in (25) and noting that $\langle \boldsymbol{r}\cdot\boldsymbol{n}, \varphi^{(i)}_{2k'+1}\rangle_{\partial T_1\cup\partial T_2} = 0$ for any i, we have

$$\langle \boldsymbol{r}\cdot\boldsymbol{n}, w_1\rangle_{\partial T_1\cup\partial T_2} = \sum_{p=0}^{2k'} \langle a^{(1)}_p \varphi^{(1)}_p, \varphi^{(1)}_{2q}\rangle_{e_1} = a^{(1)}_{2q} = 0.$$

Taking $w = \varphi^{(1)}_{2k'+1} + \varphi^{(2)}_{2q} - \varphi^{(3)}_{2k'+1}$ and $w = -\varphi^{(1)}_{2k'+1} + \varphi^{(2)}_{2k'+1} + \varphi^{(3)}_{2q}$, we get $a^{(2)}_{2q} = a^{(3)}_{2q} = 0$ for $1 \le q \le k'$. Similarly, it follows that $a^{(4)}_{2q} = a^{(5)}_{2q} = 0$ for $1 \le q \le k'$.

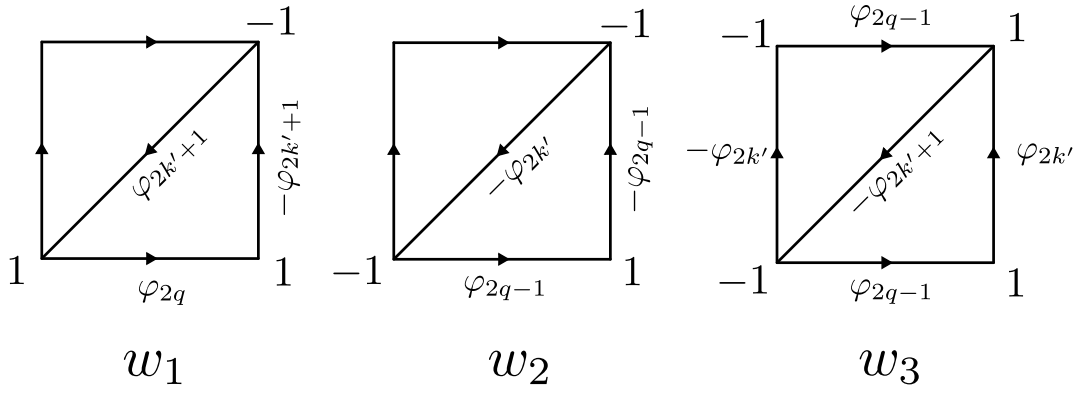


FIGURE 2. Diagrams of the test functions w_1, w_2, and w_3, where the subscripts are discarded and the number at each vertex indicates the value of the function at the vertex

Next, we show that $a^{(i)}_1 = a^{(i)}_3 = \cdots = a^{(i)}_{2k'-1}$ for $1 \le i \le 5$. In (25), choosing

$$w = w_2 := \varphi^{(1)}_{2q-1} - \varphi^{(2)}_{2q-1} + \varphi^{(3)}_{2k'},$$

in view of $a^{(i)}_{2k'} = 0$ for $1 \le i \le 5$, we have

$$\langle \boldsymbol{r}\cdot\boldsymbol{n}, w_2\rangle_{\partial T_1\cup\partial T_2} = a^{(1)}_{2q-1} - a^{(2)}_{2q-1} = 0 \quad (1 \le q \le k').$$

Choosing $w = \varphi^{(1)}_{2k'} + \varphi^{(2)}_{2q-1} - \varphi^{(3)}_{2q-1}$ in (25) yields $a^{(2)}_{2q-1} = a^{(3)}_{2q-1}$ for $1 \le q \le k'$. Similarly, we deduce that $a^{(3)}_{2q-1} = a^{(4)}_{2q-1} = a^{(5)}_{2q-1}$ for $1 \le q \le k'$. Hence, we define $a_{2q-1} := a^{(1)}_{2q-1} = \cdots = a^{(5)}_{2q-1}$ for $1 \le q \le k'$ to omit the subscripts.

Finally, we prove that $a_{2q-1} = 0$ for $1 \le q \le k'$. We take the following w_3 as a test function:

$$w_3 := \varphi^{(1)}_{2q-1} + \varphi^{(2)}_{2k'} - \varphi^{(3)}_{2k'+1} - \varphi^{(3)}_{2k'} + \varphi^{(5)}_{2q-1}.$$

Since w_3 is single valued on e_3, we see that

$$\langle \boldsymbol{r}\cdot\boldsymbol{n}, w_3|_{e_3}\rangle_{\partial T_1\cup\partial T_2} = \langle \boldsymbol{r}\cdot\boldsymbol{n}, -\varphi^{(3)}_{2k'+1}\rangle_{\partial T_1\cup\partial T_2} = 0.$$

Noting that $a^{(i)}_{2k'} = 0$ for $1 \le i \le 5$, we get

$$\langle \boldsymbol{r}\cdot\boldsymbol{n}, \varphi^{(2)}_{2k'}\rangle_{\partial T_1\cup\partial T_2} = \langle \boldsymbol{r}\cdot\boldsymbol{n}, -\varphi^{(4)}_{2k'}\rangle_{\partial T_1\cup\partial T_2} = 0$$

and

$$\langle \boldsymbol{r}\cdot\boldsymbol{n}, w_3\rangle_{\partial T_1\cup\partial T_2} = \langle \boldsymbol{r}\cdot\boldsymbol{n}, \varphi^{(1)}_{2q-1} + \varphi^{(5)}_{2q-1}\rangle_{\partial T_1\cup\partial T_2} = 2a_{2q-1} = 0.$$

Thus, we conclude that all coefficients equal zero. □

Lemma 4.2. Let K_1 and $K_2 \in \mathcal{T}_h$ be two adjacent triangles. There exists a constant C independent h such that, for $\boldsymbol{r} \in \bigoplus_{e\in\partial K_1\cup\partial K_2} \boldsymbol{N}(e)$,

$$\sum_{i=1}^{2} \|h^{1/2}\boldsymbol{r}\cdot\boldsymbol{n}\|_{L^2(\partial K_i)} \le C \sum_{i=1,2} \sup_{w\in W(K_i)} \frac{\langle \boldsymbol{r}\cdot\boldsymbol{n}, w\rangle_{\partial K_i}}{\|\nabla w\|_{L^2(K_i)} + \|h^{-1/2}\boldsymbol{P}_{\boldsymbol{N}}\nabla w\|_{L^2(\partial K_i)}}.$$

Proof. Let T_1 and T_2 be the reference triangles and let $e_1, e_2, \dots, e_5$ be the edges of T_1 and T_2. By Lemma 4.1, we see that

$$\|\boldsymbol{r}\cdot\boldsymbol{n}\|'_{\partial T_1\cup\partial T_2} := \sum_{i=1,2} \sup_{w\in W(T_i)} \frac{\langle \boldsymbol{r}\cdot\boldsymbol{n}, w\rangle_{\partial T_i}}{\|\nabla w\|_{L^2(T_i)} + \|\boldsymbol{P}_{\boldsymbol{N}} w\|_{\partial T_i}}$$

is a norm on $\bigoplus_{1\le i\le 5} \boldsymbol{N}(e_i)$. Since any two norms on a finite-dimensional space are equivalent, there exists a constant C such that

$$\|\boldsymbol{r}\cdot\boldsymbol{n}\|_{\partial T_1\cup\partial T_2} \le C\|\boldsymbol{r}\cdot\boldsymbol{n}\|'_{\partial T_1\cup\partial T_2} \quad \forall \boldsymbol{r} \in \bigoplus_{1\le i\le 5} \boldsymbol{N}(e_i).$$

By considering the Piola transforms from T_i to K_i $(i = 1, 2)$ and the scaling argument, we obtain the assertion. □

The inf-sup condition (7) immediately follows from Lemma 4.2. Similarly, we can prove the transposed inf-sup condition (8) from the following lemma.

Lemma 4.3. Let K be an element of $\mathcal{T}_h$ and $w \in W(K)$. Then, $\boldsymbol{P}_{\boldsymbol{N}} w = 0$ on ∂K if and only if

$$\langle \boldsymbol{r}\cdot\boldsymbol{n}, w\rangle_{\partial K} = 0 \qquad \forall \boldsymbol{r} \in \boldsymbol{N}(\partial K).$$

Proof. Since $\boldsymbol{P}_{\boldsymbol{N}} w \in \boldsymbol{N}(\partial K)$, we can choose $\boldsymbol{r}\cdot\boldsymbol{n} = \boldsymbol{P}_{\boldsymbol{N}} w$. We then have

$$\langle \boldsymbol{r}\cdot\boldsymbol{n}, w\rangle_{\partial K} = \langle \boldsymbol{r}\cdot\boldsymbol{n}, \boldsymbol{P}_{\boldsymbol{N}} w\rangle_{\partial K} = \|\boldsymbol{P}_{\boldsymbol{N}} w\|^2_{\partial K} = 0,$$

which implies $\boldsymbol{P}_{\boldsymbol{N}} w|_{\partial K} = 0$. □

The proof of the transposed inf-sup condition is the same as in Lemma 4.1, so we omit it here.

Remark 4.4. We have proved that the inf-sup condition (7) holds for the triangular elements where $\boldsymbol{N}_h$ and W_h are polynomials of degree k and $k+1$, respectively. However, it is still open whether there exists a pair of $\boldsymbol{N}_h$ and W_h satisfying the inf-sup condition (7) in the three- or higher-dimensional cases.

5. Numerical results

In this section, we examine the convergence property of the proposed method (5) by numerical experiments. As a test problem, we consider the Poisson equation with homogeneous Dirichlet boundary condition

$$\begin{aligned} -\Delta u &= 2\pi^2 \sin(\pi x)\sin(\pi y) && \text{in } \Omega, \\ u &= 0 && \text{on } \partial\Omega, \end{aligned}$$

where $\Omega = (0,1)^2$ and the exact solution is given by $u(x,y) = \sin(\pi x)\sin(\pi y)$. We use the unstructured triangulations whose mesh sizes are approximately 0.4×2^{-i} $(1 \le i \le 4)$ and compute the solution of (5), varying the polynomial degree k from 0 to 3. All numerical computations are carried out by FreeFEM [4]. The L^2-errors of $\boldsymbol{q}_h$ and u_h are displayed in Table 5. We observe that the orders of convergence in $\boldsymbol{q}_h$ and u_h are $k+1$ and $k+2$, respectively, which are of optimal order and fully agrees with Theorems 3.2 and 3.3.

Table 1. Convergence history for the method (5)

k	h	$\|\boldsymbol{q} - \boldsymbol{q}_h\|$	Order	$\|u - u_h\|$	Order
	0.1901	3.472E-01	—	1.027E-02	—
0	0.1025	1.660E-01	1.20	2.404E-03	2.35
	0.0509	8.355E-02	0.98	6.150E-04	1.94
	0.0262	4.120E-02	1.07	1.461E-04	2.17
	0.1901	1.827E-02	—	2.868E-04	—
1	0.1025	5.218E-03	2.03	4.389E-05	3.04
	0.0509	1.221E-03	2.07	4.926E-06	3.12
	0.0262	2.967E-04	2.14	5.872E-07	3.21
	0.1901	1.172E-03	—	1.537E-05	—
2	0.1025	1.212E-04	3.68	7.598E-07	4.87
	0.0509	1.465E-05	3.01	4.524E-08	4.02
	0.0262	1.797E-06	3.17	2.732E-09	4.24
	0.1901	3.283E-05	—	2.773E-07	—
3	0.1025	2.662E-06	4.07	1.185E-08	5.11
	0.0509	1.408E-07	4.19	3.021E-10	5.23
	0.0262	8.376E-09	4.27	1.326E-11	4.73

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